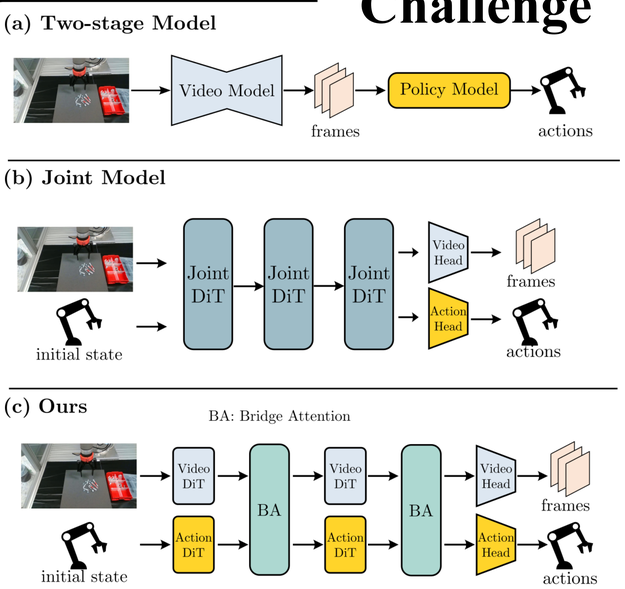
CoVAR: Co-generation of Video and Action for Robotic Manipulation via Multi-Modal Diffusion

Liudi Yang, Yang Bai, George Eskandar, Fengyi Shen, Mohammad Altillawi,
Dong Chen, Ziyuan Liu, and Abhinav Valada

ICRA IEEE INTERNATIONAL CONFERENCE ON ROBOTICS AND AUTOMATION
2026 VIENNA

Challenge

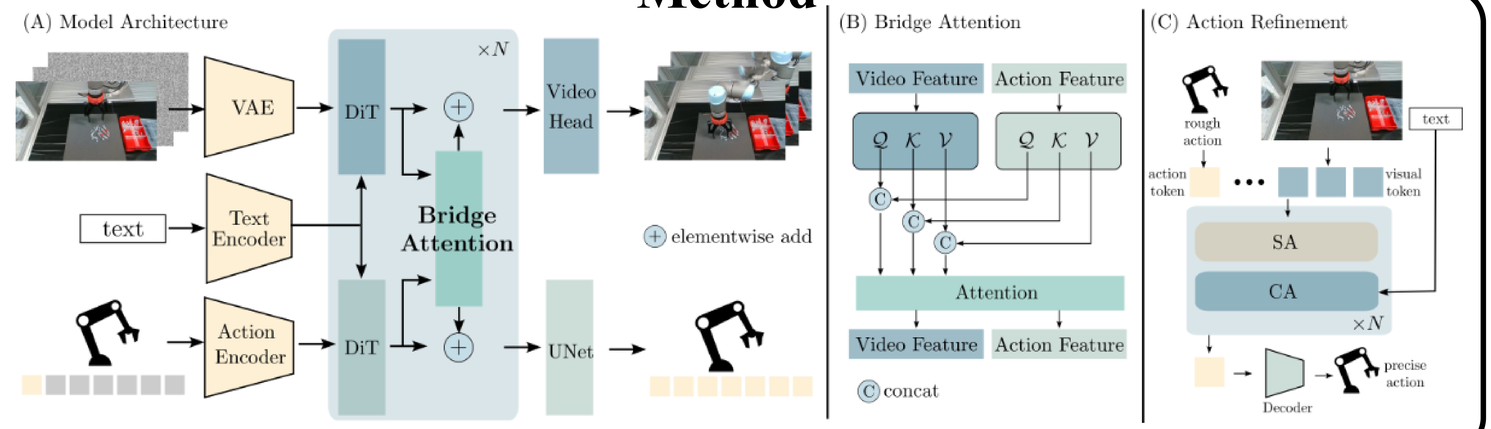


Two-stage Model:
Strong dependence on video generation quality

Joint Model:
Require large-scale adaptation of video diffusion models to joint video-action space

Ours:
Parallel action DiT enables joint generation while preserving pretrained knowledge

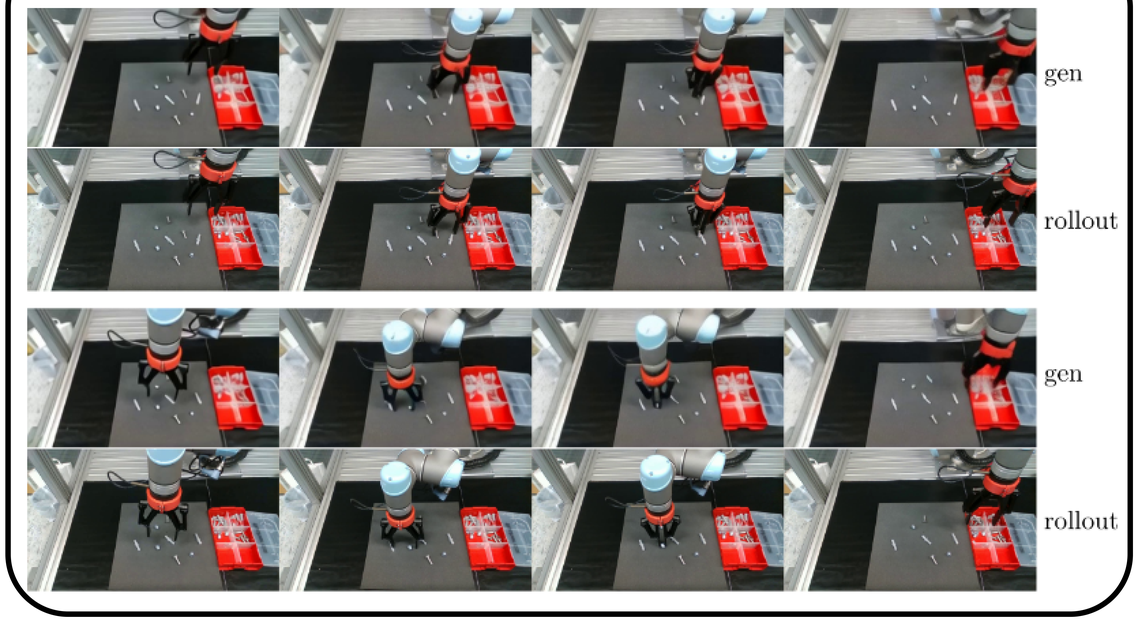
Method



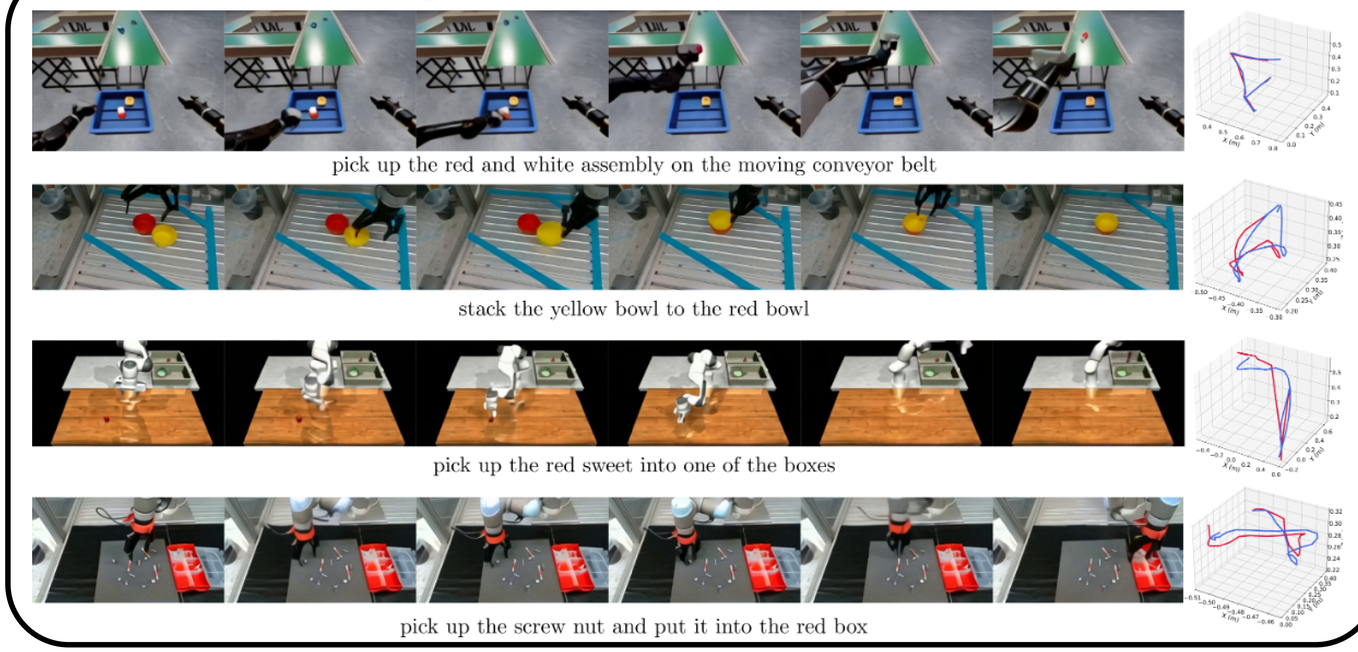
Contributions

1. Unified video–action generation via parallel action DiT.
2. Improved cross-modal alignment with Bridge Attention.
3. Higher-precision actions through refinement and stronger real-world performance.

Generation vs. Rollout



Quantitative Results



Qualitative Results

Video Quality

	Calvin				Libero90			
	PSNR	SSIM	LPIPS	FVD	PSNR	SSIM	LPIPS	FVD
UVA [8]	19.01	0.758	0.180	97.90	19.57	0.716	0.154	86.21
PAD [7]	18.72	0.734	0.174	83.40	19.65	0.781	0.218	98.39
UWM [9]	18.04	0.730	0.181	85.85	19.87	0.735	0.212	87.83
OpenSora [28]	19.60	**0.768**	0.171	**61.00**	**20.18**	0.817	0.156	**63.33**
Ours	**19.95**	0.766	**0.156**	72.42	20.09	**0.826**	**0.143**	70.64

Success Rate

Real	Nut	Screw	Dowel
Unipi [4]	0.00	0.06	0.02
RoboEnvision [31]	0.04	0.10	0.12
Ours	**0.64**	**0.74**	**0.70**

Calvin	Drawer	Cabinet	Light	Pick	Push
UVA [8]	0.875	0.667	0.711	0.758	0.785
UWM [9]	0.813	0.733	0.644	0.576	0.714
PAD [7]	0.781	0.467	0.489	0.485	0.642
Unipi [4]	0.469	0.267	0.289	0.182	0.452
Ours	**1.000**	**0.800**	**0.867**	**0.909**	**0.929**